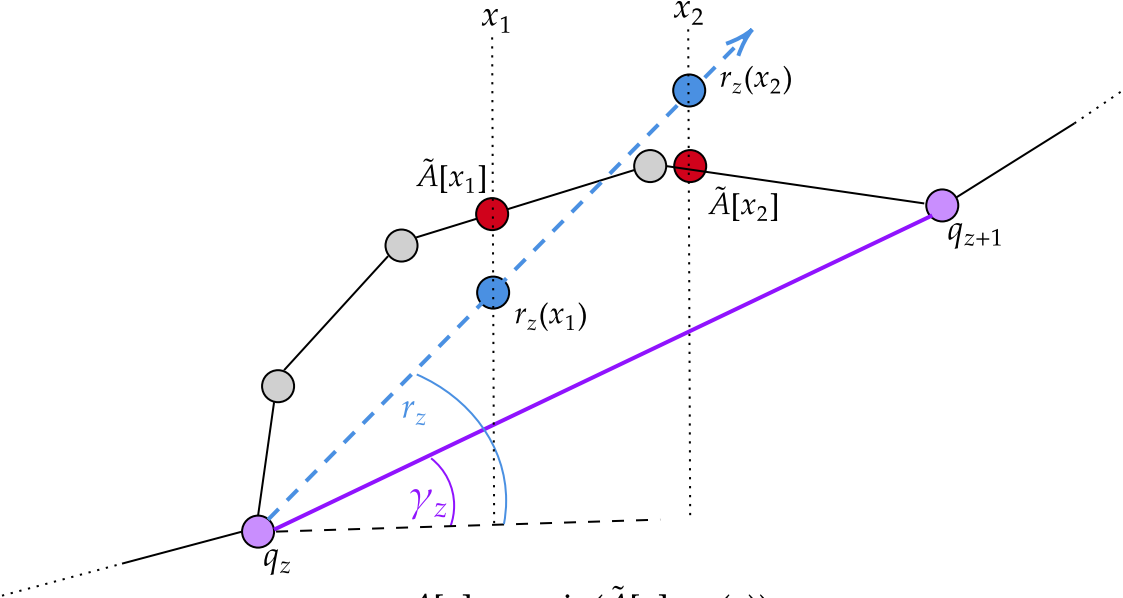

$$A[x] \;=\; \min(\tilde{A}[x],\; r_z(x))$$